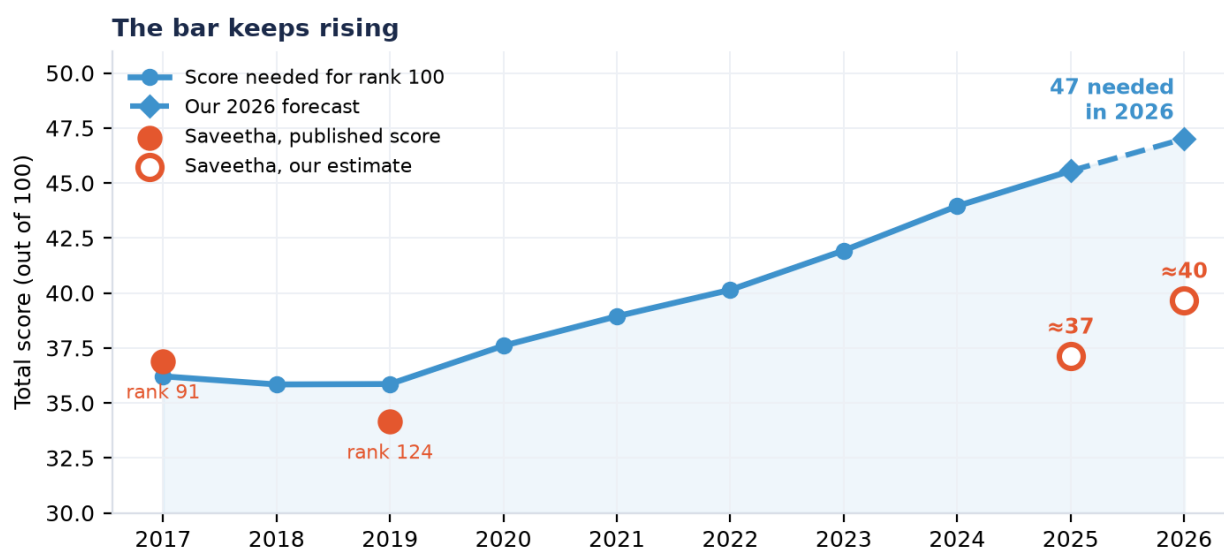
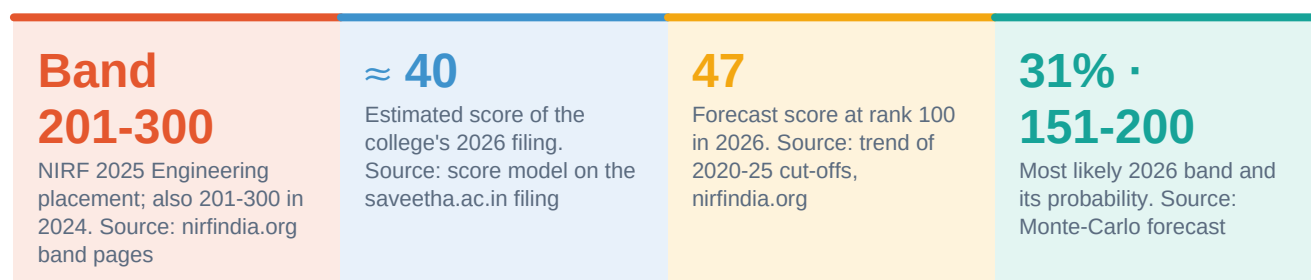


NIRF Analytics

Position, drivers and the 2026 outlook

An evidence-based reading of where the college stands in the NIRF Engineering rankings, how the score is built, and what the data says about 2026.

Prepared 09 September 2026 · Sources: nirfindia.org 2017-2025, NIRF methodology PDFs, the college's 2025 and 2026 filings
Every figure carries its source in grey. Grey boxes hold technical detail and can be skipped.



The blue line is the lowest score inside the top 100 each year. The solid red dots are the college's two published scores; the hollow ones are model estimates for its last two filings.

Sources: S1 NIRF ranking tables (cut-off = score at rank 100; the college's 2017 and 2019 rows); S6/S7 the college's 2025 and 2026 filings; S8 score model. Full list in section 12.

1 Summary of findings

Saveetha Engineering College was ranked **91 in 2017** (score 36.88) and **124 in 2019** (score 34.14) in NIRF's Engineering category. [NIRF ranking tables, 2017 and 2019 tables] In 2024 and 2025 it was placed in the **201-300 band**. [NIRF band pages, 2024, 2025] NIRF publishes no scores for band placements, so the college's actual parameter scores for those years are not public; the estimates in this report come from a model described in section 6.

The cut-off has risen every year since 2019. The lowest score inside the top 100 was 41.93 in 2023, 43.95 in 2024 and 45.55 in 2025. [NIRF ranking tables, score at rank 100] A linear trend on 2020-2025 gives about 47.0 for 2026. [2026 forecast, thresholds] The college's filings are estimated at 37.1 (2025) and 39.7 (2026). [score model, saveetha_estimates] The 2025 estimate falls inside the band NIRF actually published, which is one check on the model.

The gap is concentrated in research. The estimated Research and Professional Practice score (RPC) for the 2026 filing is 13.4; institutes ranked 76-100 in 2025 average 29.8. [analysis.json, tier_profile_2025] In the filings themselves: 12 PhDs awarded per year against a tier median of 24; 61 full-time PhD scholars against 163; about Rs 30 lakh a year of sponsored research against Rs 3.7 crore. [2026 filing] [institute PDFs, 2025 top-100 medians] Placement (97%), on-time graduation (81%) and women students (33%) are at or above the 76-100 tier medians (70%, 87%, 25%). [2026 filing] [analysis.json, raw_profile_2025]

2026 outlook. The most likely outcome is the **151-200 band** (31% of simulations); top 100 in 4% of simulations; top 200 in 58%. [2026 forecast, band_probabilities] The range is wide because two inputs are unobservable (section 8).

LEVERS WITH THE LARGEST ESTIMATED EFFECT [2026 FORECAST, WHAT_IF_LEVERS]

Ranked by the model's estimate of value: PhDs awarded (12 → 80 a year: +3.7 points), operating spend per student (Rs 1.1 L → 2 L: +2.0), full-time PhD scholars (61 → 200: +1.4), median UG salary (Rs 5.5 L → 10 L: +1.4), sponsored research (Rs 30 L → 10 Cr a year: +1.0). Each figure is one change applied alone to the 2026 filing; publications and citations are not in the data and are not counted here.

What this report is based on. Every NIRF Engineering result since 2017, the raw data PDFs of 704 institute submissions, NIRF's methodology documents for 2023-25, the college's own 2025 and 2026 filings, a model fitted on 700 institute-years, and a 2026 forecast. Section 12 lists each source with its location.

2 How NIRF scores an engineering institution

NIRF is the Ministry of Education's annual ranking, run since 2016 with the National Board of Accreditation as ranking agency. [NIRF 2025 methodology, section 4.1] Institutions apply voluntarily and upload their own data; publication, citation and patent counts come from third-party databases. [NIRF 2025 methodology, section 3.5] Each institution receives a score out of 100 from five parameters with fixed weights: [NIRF 2025 methodology, summary table]

$$\text{Total} = 0.30 \times \text{TLR} + 0.30 \times \text{RPC} + 0.20 \times \text{GO} + 0.10 \times \text{OI} + 0.10 \times \text{PR}$$

Parameter	Sub-parameter	Marks	In plain words
TLR · Teaching, Learning & Resources (weight 0.30)	Student Strength including Doctoral Students (SS)	20	Number of students, incl. PhD scholars
TLR · Teaching, Learning & Resources (weight 0.30)	Faculty-student ratio with emphasis on permanent faculty (FSR)	30	Faculty-student ratio; 1:15 earns full marks
TLR · Teaching, Learning & Resources (weight 0.30)	Combined metric for Faculty with PhD (or equivalent) and Experience (FQE)	20	Share of faculty with a PhD, and the experience mix
TLR · Teaching, Learning & Resources (weight 0.30)	Financial Resources and their Utilisation (FRU)	30	Money spent per student, capital and operating
RPC · Research & Professional Practice (weight 0.30)	Combined metric for Publications (PU)	35	Publications per faculty (Scopus / Web of Science)
RPC · Research & Professional Practice (weight 0.30)	Combined metric for Quality of Publications (QP)	40	Citations and top-25% papers per faculty
RPC · Research & Professional Practice (weight 0.30)	IPR and Patents: Published and Granted (IPR)	15	Patents published and granted
RPC · Research & Professional Practice (weight 0.30)	Footprint of Projects and Professional Practice (FPPP)	10	Research funding and consultancy income
GO · Graduation Outcomes (weight 0.20)	Combined metric for Placement and Higher Studies (GPH)	40	Share placed plus share going to higher studies
GO · Graduation Outcomes (weight 0.20)	Metric for University Examinations (GUE)	15	Share graduating on time; 80% earns full marks
GO · Graduation Outcomes (weight 0.20)	Median Salary (GMS)	25	Median salary of placed graduates
GO · Graduation Outcomes (weight 0.20)	Metric for Number of Ph.D. Students Graduated (GPHD)	20	PhDs awarded per year
OI · Outreach & Inclusivity (weight 0.10)	Economically and Socially Challenged Students (ESCS)	20	Students on full fee reimbursement
OI · Outreach & Inclusivity (weight 0.10)	Facilities for Physically Challenged Students (PCS)	20	Facilities for physically challenged students
PR · Perception (weight 0.10)	Peer Perception: Employers & Academic Peer (PR)	100	A survey of employers and academics

Source: NIRF methodology PDF, Engineering 2025 (nirfindia.org/nirfpdfcdn/2025/framework/Engineering.pdf). Marks and weights as printed in the 'Summary of Ranking Parameters and Weightages – 2025 (Engineering)' table; plain-words column is our paraphrase.

TECHNICAL NOTE — SKIP IF YOU LIKE

Formulas as printed in the 2025 Engineering methodology. $FSR = 30 \times [15 \times (F/N)]$, F = full-time faculty, N = students incl. PhD scholars; zero below 1:50. $FQE = 10 \times (FRA/95)$ for the PhD share plus up to 10 for an even spread over ≤ 8 , 8-15 and >15 years of experience. $FRU = 7.5 \times f(BC) + 22.5 \times f(BO)$ on capital and operating spend per student. $PU = 35 \times f(P/FRQ) - 5 \times f(Pret)$; $QP = 20 \times f(CC/FRQ) + 20 \times f(TOP25P/P) - 5 \times f(Cret)$; the -5 terms for retracted work are new in 2025. $IPR = 10 \times f(\text{patents granted}) + 5 \times f(\text{patents published})$. $GPH = 40 \times (Np/100 + Nhs/100)$. $GUE = 15 \times \min(Ng/80, 1)$. $GMS = 25 \times f(\text{median salary})$. $GPHD = 20 \times f(\text{PhDs graduated})$. $RD = 25 \times (\text{share from other states}) + 5 \times (\text{share from abroad})$. $WD = 15 \times (\text{women students \%}/50) + 15 \times (\text{women faculty \%}/20)$. $ESCS = 20 \times f(\text{fee-reimbursed \%})$. The $f()$ functions are described in the document as 'to be determined by NIRF' and are not published.

TWO POINTS OF DEFINITION

Only the top 100 receive a numeric rank; institutions below are listed in bands (101-150, 151-200, 201-300) without scores [NIRF band pages]. **Saveetha Engineering College** (Sriperumbudur, NIRF id IR-E-C-16590) and **Saveetha Institute of Medical and Technical Sciences** (Chennai, IR-E-I-1441, rank 45 in 2025) are separate NIRF entities [NIRF 2025 Engineering table; the college's filing header]. This report covers the college only.

3 The college's NIRF record

Year	Category	Position	Total	TLR	RPC	GO	OI	PR
2017	Engineering	Rank 91	36.88	49.94	4.73	64.45	70.27	5.64
2018	Engineering	Band 101-150	—	—	—	—	—	—
2018	Overall	Band 151-200	—	—	—	—	—	—
2019	Engineering	Rank 124	34.14	55.68	2.99	55.82	53.45	0.41
2019	Overall	Band 151-200	—	—	—	—	—	—
2020	Engineering	Band 201-250	—	—	—	—	—	—
2024	Engineering	Band 201-300	—	—	—	—	—	—
2025	Engineering	Band 201-300	—	—	—	—	—	—

Source: nirfindia.org → Rankings → Engineering, yearly tables (2017-2025); nirfindia.org → Rankings → Engineering → Rank-band pages. '—' = no score published (band placement). 2021-2023: the college appears in the participant lists but not in the ranked lists or bands. In 2020 NIRF published 200 numeric ranks, so the page labelled '101-150' holds 201-250; the table reflects that.

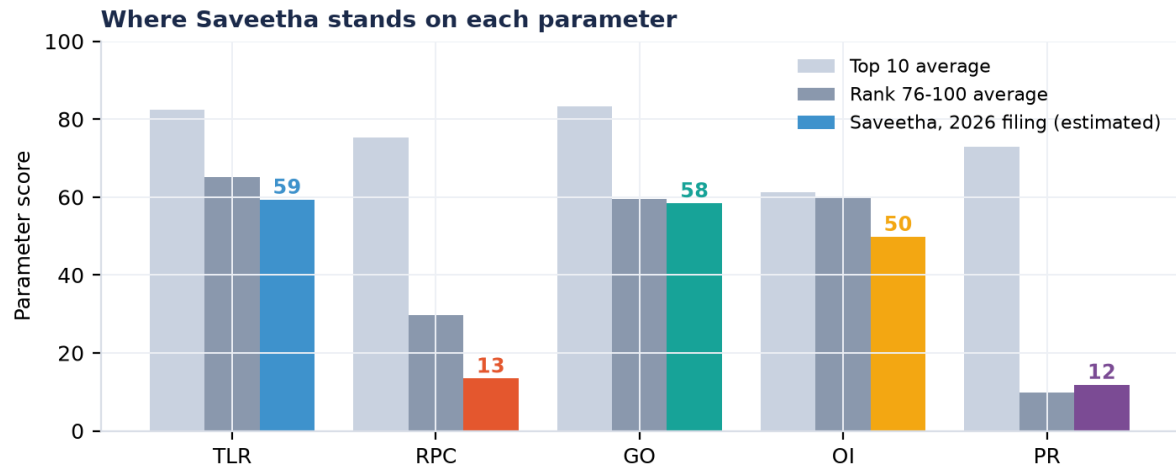
What the last two filings are worth, by the model

Filing	TLR	RPC	GO	OI	PR (assumed)	Estimated total	Rank-100 score that year
2025	55.5	11.6	52.9	52.3	11.9	37.1	45.55
2026	59.3	13.4	58.4	49.8	11.9	39.7	≈ 47.0 (forecast)

Source: Score model output, models/score_model.py → data/processed/model_report.json; College's NIRF 2025 Engineering filing, saveetha.ac.in/nirf-documents-2024 (Engineering-1.pdf); College's NIRF 2026 Engineering filing, saveetha.ac.in (SAVEETHA-ENGINEERING-COLLEGE20260216-.pdf). PR is not in the filing; the value shown is the median Perception score of private colleges ranked 60-100 in 2025 [NIRF ranking tables].

The 2025 filing's estimate (37.1) lies within the 201-300 band that NIRF announced for the college in 2025; the model was not given that information. Between the two filings the reported numbers moved: on-time graduation 61% → 81%, placement 92% → 97%, operating spend per student Rs 80,210 → Rs 1,10,595,

PhDs awarded 9 → 12 a year. [2025 filing] [2026 filing]



Estimated parameters for the college's 2026 filing (coloured) beside the 2025 averages of institutes ranked 76-100 and 1-10.

Source: Score model output, `models/score_model.py` → `data/processed/model_report.json`; `analysis/analyze.py` → `data/processed/analysis.json` (computed from `db/nirf.db`). Tier averages from `tier_profile_2025`.

4 Data collected

Source	What it gives	Size	Where
NIRF ranking pages	Engineering, Overall, College and University lists 2017-2025: rank, total and the five parameter scores (Engineering ranks 1-200 for 2019-22)	4,001 rows	nirfindia.org/Rankings//EngineeringRanking.html
NIRF band pages	Institutions placed in 101-150 / 151-200 / 201-300, no scores	3,852 rows	.../EngineeringRanking150.html, 200, 300
NIRF participant lists	Every institution that applied, per year and category	47,173 rows	.../EngineeringRankingALL.html
Per-institute data PDFs	Each ranked institute's submitted data: intake, enrolment by gender/state/category, placements, median salary, higher studies, PhDs, capital and operating spend, sponsored research, consultancy, patents, facilities. Top 100 for 2023-25; top 200 for 2021-22	704 PDFs	nirfindia.org/nirfpdfcdn//pdf/Engineering/.pdf
NIRF methodology PDFs	Definitions, marks and formulas for 2023, 2024, 2025 (Engineering, Overall, College)	123 rows	nirfindia.org/nirfpdfcdn//framework/.pdf
College's own filings	NIRF 2025 Engineering, Innovation and SDG filings and the NIRF 2026 Engineering filing, incl. the full faculty list	4 PDFs · 1,575 faculty rows	saveetha.ac.in/nirf-documents-2024/
Full text of the above	Indexed for the AI assistant	713 documents	db/nirf.db, table documents

Source: `db/nirf.db` (all of the above, loaded by `scraper/build_db.py`). Row counts are live from the database at build time.

TECHNICAL NOTE — SKIP IF YOU LIKE

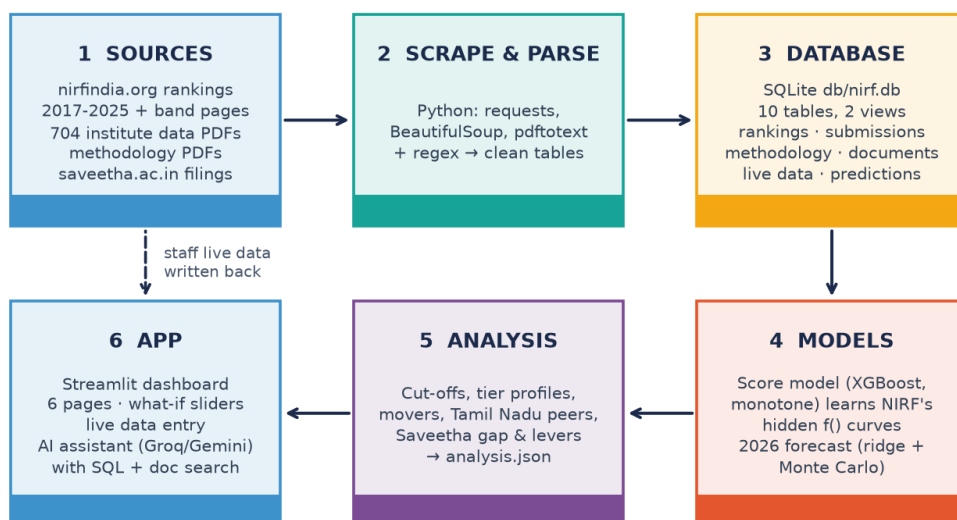
Data handling. The 2016 pages on nirfindia.org return the 2017 tables (identical content, title 'India Rankings 2017'), so 2016 is dropped. In years with 200 numeric ranks the pages labelled '101-150' and '151-200' hold ranks 201-250 and 251-300 (no overlap with the ranked list), and are relabelled. Band pages carry no institute IDs, so band rows are linked to institutions by a normalised name key. PDFs are converted with `pdftotext -layout` and parsed with regular expressions; every numeric field the model uses has over 96% coverage across the 704 PDFs. Faculty lists appear only in the college's own PDFs.

The college's raw numbers beside the 2025 top 100

Metric	Top 10 (median)	Rank 76-100 (median)	Colleges in top 100 (median)	College, 2025 filing	College, 2026 filing
Students (UG+PG)	6,442.5	4,040	5,242	5,196	5,961
Faculty	590	237	404	—	—
Students per faculty	14.7	14.6	13.0	13.5	14.9
Full-time PhD scholars	2,107	163	232	54	61
PhDs awarded / year (3-yr avg)	235.3	24	44.7	9	12
Graduating on time	81%	87%	89%	61%	81%
Placed	77%	70%	77%	92%	97%
Median UG salary	Rs 19.44 L	Rs 7.28 L	Rs 7.00 L	Rs 5.20 L	Rs 5.50 L
Capital spend / student (3-yr avg)	Rs 2.22 L	Rs 48,944	Rs 31,954	Rs 15,666	Rs 18,299
Operating spend / student (3-yr avg)	Rs 10.07 L	Rs 2.66 L	Rs 2.04 L	Rs 80,210	Rs 1.11 L
Sponsored research / yr (3-yr avg)	Rs 154.87 Cr	Rs 3.66 Cr	Rs 2.06 Cr	Rs 25.24 L	Rs 29.83 L
Consultancy / yr (3-yr avg)	Rs 43.61 Cr	Rs 2.01 Cr	Rs 1.28 Cr	Rs 34.15 L	Rs 20.04 L
Women students	20%	25%	36%	31%	33%
Students from other states	79%	57%	35%	10%	8%

Source: nirfindia.org per-institute data PDFs (nirfpdfcdn/pdf/Engineering/.pdf); College's NIRF 2025 Engineering filing, saveetha.ac.in/nirf-documents-2024 (Engineering-1.pdf); College's NIRF 2026 Engineering filing, saveetha.ac.in (SAVEETHA-ENGINEERING-COLLEGE20260216-.pdf); analysis/analyze.py → data/processed/analysis.json (computed from db/nirf.db). Red = 2026 filing below the rank 76-100 median; green = at or above it. Medians over the 100 institutes ranked in 2025.

5 How the system works



One command (`./pipeline.sh`) re-runs every step when NIRF 2026 is published

Scripts fetch the public data (1) and turn web tables and PDFs into clean rows (2), which are stored in one database file (3). The models (4) are fitted on the 700 institute-years where both the raw submission and the published score are known, then applied to the college's filings. The analysis step (5) produces the comparisons in this report, and the app (6) presents them, accepts live numbers from staff, and answers questions through an AI assistant that shows the queries it runs.

TECHNICAL NOTE — SKIP IF YOU LIKE

Stack: Python 3.13; requests + BeautifulSoup for scraping; poppler pdftotext + regex for PDF parsing; pandas; SQLite; XGBoost and scikit-learn for models; Streamlit + Plotly for the app; rank-bm25 for document retrieval; Groq (gpt-oss-120b) as the chat model with Gemini as fallback, both on free tiers. Layout: scraper/ (4 scripts) · models/ (score_model.py, predict_2026.py, artifacts/) · analysis/analyze.py · app/ (Home.py, common.py, pages/1-6, rag/) · data/raw (html, pdf, pdf_text, methodology, saveetha) · data/processed (CSV + JSON) · db/nirf.db · docs/.

6 The scoring model

Why a model is needed. NIRF publishes the formula but not the normalisation functions $f()$ that turn a raw number into marks. [NIRF 2025 methodology, e.g. SS: f to be determined by NIRF] Without them a score cannot be computed from raw data.

Approach. For 700 institute-years (Engineering, ranks 1-200 in 2021-22 and 1-100 in 2023-25) both sides are known: the raw submission and the parameter scores NIRF published. One model per parameter is fitted on exactly the inputs the methodology lists for that parameter, then applied to the college's filings. [score model, features per parameter]

Choice of method. Gradient-boosted trees (XGBoost) with monotone constraints. Trees capture the saturating shape the methodology implies (a ratio of 1:15 earns full FSR marks; more does not add). Monotone constraints require the fitted function to move in the direction the methodology states (more placements cannot reduce GO; more spend cannot reduce TLR), which removes spurious patterns and keeps what-if results consistent. A linear model on the same inputs reached R^2 of about 0.5 in our tests; a

neural network would need far more rows.

Parameter	Rows	Cross-validated R ²	Typical error (MAE)	Largest contributors (gain share)
TLR	700	0.73	4.3 pts	phd pursuing ft (36%), log opex per student (21%), faculty per student (12%)
RPC	700	0.78	7.5 pts	phd pursuing ft (60%), log sponsored amount (20%), phd grad 3y avg (16%)
GO	700	0.82	3.7 pts	log median salary (49%), phd grad 3y avg (26%), placed or hs rate (18%)
OI	700	0.74	3.0 pts	outside state pct (41%), women students pct (27%), pcs score 0 3 (17%)

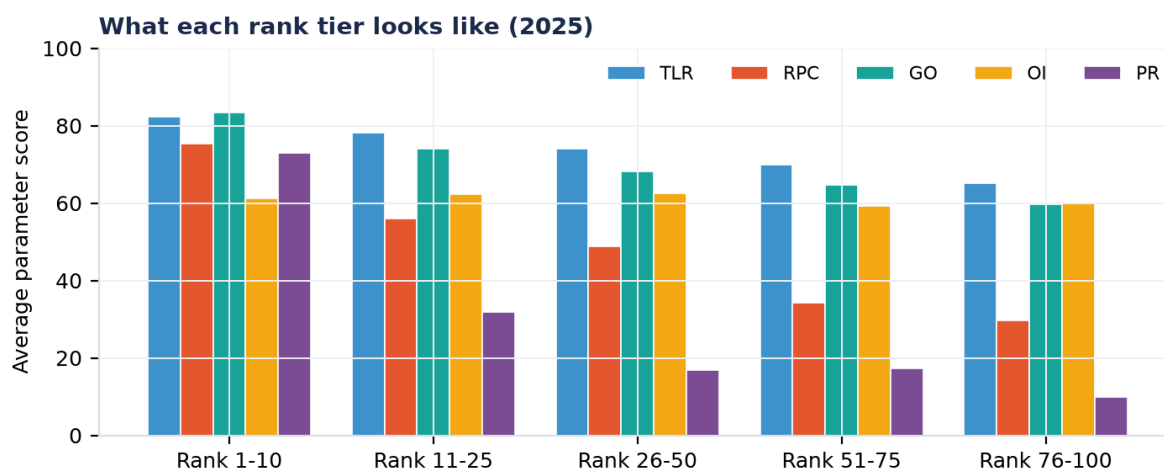
Source: Score model output, `models/score_model.py` → `data/processed/model_report.json`. 5-fold cross-validation grouped by institute, so an institute's other years are never in its training fold.

With the published Perception score added, the four estimates reproduce published totals with a mean absolute error of **2.05 points** (R² 0.952) on the training rows. [`score model`, `total_score_fit`]

KNOWN LIMITS [MODEL_REPORT.JSON, PREDICTION_2026.JSON]

Publications and citations carry 75 of RPC's 100 marks and are not in the submission PDFs, so RPC is inferred from PhD output, funding and faculty size; it has the largest error. Perception is a survey and cannot be modelled from data; the forecast treats it as an uncertainty drawn from private colleges ranked 60-100 in 2025 (median 11.9, 10th-90th percentile 3.8-48.8). The model is trained on institutes ranked 1-200 and is most reliable in that range, which is where the college's 2026 filing is estimated to sit.

7 The 2025 field



RPC and Perception rise steeply with rank; TLR, GO and OI are flatter. The 2025 top 100 contained 87 universities, 9 colleges and 4 deemed or national institutes; colleges numbered 12 in 2023 and 9 in 2025.

Source: nirfindia.org → Rankings → Engineering, yearly tables (2017-2025); analysis/analyze.py → data/processed/analysis.json (computed from db/nirf.db). Type from the NIRF id (U / C / I); tier averages from tier_profile_2025.

Rank	Tamil Nadu institutions in the 2025 Engineering top 100	Type	Total	RPC
1	Indian Institute of Technology Madras	University	88.72	90.7
9	National Institute of Technology Tiruchirappalli	University	68.14	57.6
14	S.R.M. Institute of Science and Technology	University	65.83	68.9
16	Vellore Institute of Technology	University	65.25	67.6
20	Anna University	University	63.51	62.8
23	Amrita Vishwa Vidyapeetham	University	62.46	51.8
33	Kalasalingam Academy of Research and Education	University	59.03	42.5
40	Shanmugha Arts Science Technology & Research Academy	University	58.02	35.9
45	Saveetha Institute of Medical and Technical Sciences	Institute	56.55	61.6
47	Sri Sivasubramaniya Nadar College of Engineering	College	56.08	44.0
67	Sathyabama Institute of Science and Technology	University	50.64	36.4
67	PSG College of Technology	College	50.64	26.3
87	Vel Tech Rangarajan Dr. Sagunthala R & D Institute of Science and Technology	University	47.43	35.8
100	Sri Krishna College of Engineering and Technology	College	45.55	22.4

Source: nirfindia.org → Rankings → Engineering, yearly tables (2017-2025). 2025 Engineering table filtered to state = Tamil Nadu. Colleges (id IR-E-C-...) in red.

Three of the fourteen are colleges: SSN (rank 47, RPC 44.0), PSG Tech (67, RPC 26.3) and Sri Krishna CET (100, RPC 22.4). Their RPC values bracket what a college at the cut-off carries.

Where the weighted points are

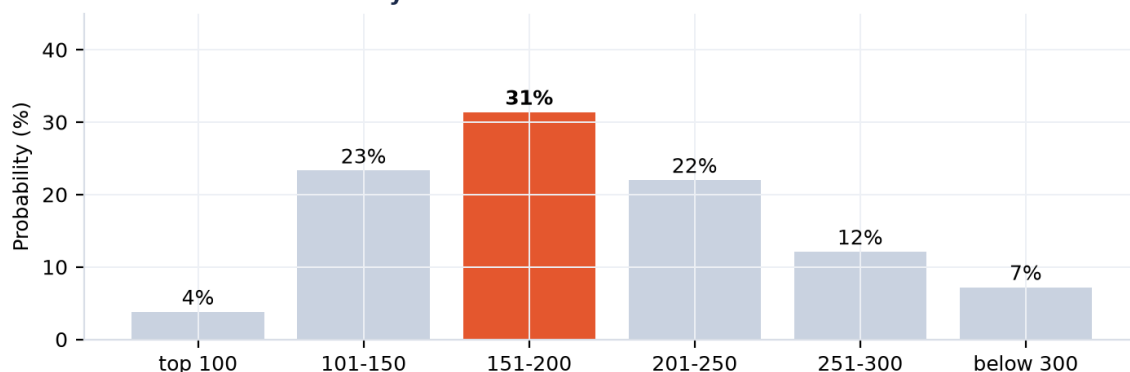
Parameter	Rank 90-100 average (2025)	College, 2026 filing (est.)	Difference	× weight = points
TLR	64.5	59.3	+5.2	+1.56
RPC	27.1	13.4	+13.7	+4.11
GO	60.7	58.4	+2.3	+0.46
OI	57.6	49.8	+7.8	+0.78
PR	10.8	11.9	-1.1	-0.11

Source: analysis/analyze.py → data/processed/analysis.json (computed from db/nirf.db); Score model output, models/score_model.py → data/processed/model_report.json. saveetha_gap_vs_rank90_100_avg and saveetha_weighted_gap_by_param; weights from the 2025 methodology.

RPC accounts for 4.1 of the 6.9 weighted points between the college's estimate and the rank 90-100 average; TLR and OI (students from other states) account for most of the rest; GO is close to level.

8 The 2026 forecast

Where Saveetha is likely to land in 2026



40.12

Median estimated total;
10th-90th percentile
35.61-45.15

≈ rank 187

Median total placed on the
2026 threshold curve

58%

Share of simulations inside
the top 200

6.9 pts

Median total vs forecast
rank-100 score (47.0)

Rank in 2026	100	125	150	175	200	250	300
Forecast score at that rank	47.0	44.4	42.5	40.9	39.3	37.0	35.0

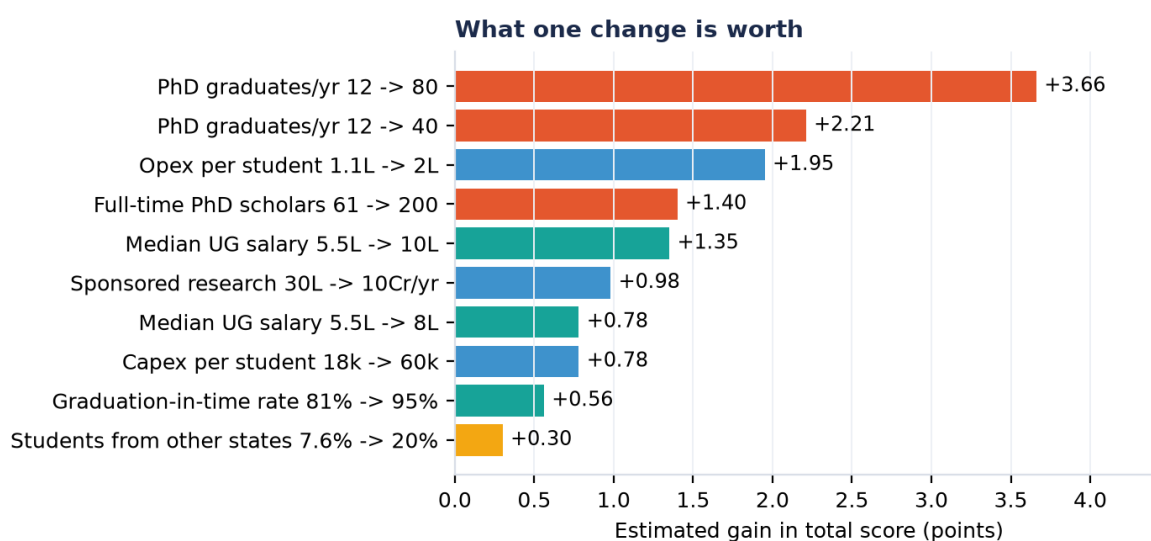
Source: Forecast output, models/predict_2026.py → data/processed/prediction_2026.json. thresholds.forecast_2026.

TECHNICAL NOTE — SKIP IF YOU LIKE

Method. (a) Thresholds: linear trend of the rank-100 score over 2020-2025 (+1.5 a year, residual s.d. 0.5); the ratio of the score at rank k to the score at rank 100 is taken from 2019-22, when NIRF published 200 ranks, and extrapolated to 300 on a log-rank scale. (b) The college: parameter estimates from the score model on the 2026 filing; 20,000 Monte-Carlo draws add each parameter model's cross-validated RMSE, a Perception score sampled from private colleges ranked 60-100 in 2025, and the threshold trend's residual; each draw is assigned a band. (c) Other institutions: a ridge regression on previous score, parameters and two-year momentum predicts each 2025 top-100 institute's 2026 score. Trained on ≤ 2024 and tested on 2025 it has MAE 1.36 against 1.62 for a no-change assumption, and Spearman 0.963 against the actual 2025 order [prediction_2026.json, top100_model_validation].

Reading the probabilities. The spread comes mainly from two inputs the data does not contain: the Perception survey and the publication counts behind RPC. Strong Scopus output would move the college's true score towards the upper end of the range; the model cannot see it either way.

9 What moves the score



Each bar is one change applied alone to the 2026 filing with everything else unchanged, scored through the same model. Effects are approximately additive.

Source: Forecast output, `models/predict_2026.py` → `data/processed/prediction_2026.json`. `what_if_levers`; each lever re-scores the filing with a single field changed.

Observations, with the methodology reference for each

- **PhDs awarded** is the largest single lever: 12 → 40 a year is worth about +2.2, 12 → 80 about +3.7. GPHD carries 20 marks of GO; PhD scholars also count in SS (TLR), and PhD output is the strongest predictor of RPC in the model (60% of gain). Awards lag enrolment by three to four years; 61 → 200 full-time scholars is worth +1.4 on its own.
- **Publications and citations** carry 75 of RPC's 100 marks [methodology] and are not observed here. Institutes ranked 76-100 average RPC 29.8 against the college's estimated 13.4. NIRF 2025 introduced a deduction for retracted papers ($-5 \times f(\text{Pret})$, $-5 \times f(\text{Cret})$).
- **Operating and capital spend per student**: FRU is 30 marks of TLR. Rs 1.1 L → Rs 2 L operating spend per student is worth about +2.0; Rs 18k → Rs 60k capital spend about +0.8. The methodology excludes buildings from capital spend and hostels from operating spend.
- **Median UG salary**: GMS is 25 marks of GO. Rs 5.5 L → Rs 8 L is +0.8; → Rs 10 L is +1.4. The metric is the median of placed graduates, not the highest offer.
- **Sponsored research and consultancy**: FPPP is 10 marks of RPC. Rs 30 L → Rs 10 Cr a year of sponsored research is about +1.0 in the model.
- **Students from other states**: RD is 30 marks of OI. The college reports 7.6% against a tier median of 57%; 7.6% → 20% is worth about +0.3.
- **Already at or above the tier** in the 2026 filing: placement 97%, on-time graduation 81%, women students 33%, full fee reimbursement 52%.

PUTTING THE NUMBERS TOGETHER [PREDICTION_2026.JSON]

The difference between the college's median estimate (40.12) and the forecast rank-100 score (47.0) is 6.9 points, and the rank-100 score has been rising about 1.5 points a year. Combining the levers above at the mid-range values (PhDs → 40 a year, operating spend → Rs 2 L, salary → Rs 8 L, scholars → 200) adds an estimated 6-8 points by the model, before any effect from publications.

10 The app, and where things live

Page	What it shows
Home	The college's band, the cut-off trend, the 2026 forecast, and how the top 100 earns its points
Top 100 Explorer	Every ranked Engineering institute 2017-2025: filter by year, state, type; parameter scores, the weighted 'score anatomy', the raw numbers each institute submitted, and any institute's history
Saveetha Position	The college's NIRF history, estimated parameter scores, the gap table, raw numbers against the tiers, and the paths of Tamil Nadu colleges in the top 100
Gap & What-If	A slider for every lever. The model re-scores instantly and shows the implied 2026 band
Prediction 2026	Forecast thresholds, band probabilities, model accuracy, and the projected 2026 order of the 2025 top 100
Live Data Entry	Staff enter current-year numbers (placements, salary, PhDs, spend, Scopus counts). Entries are time-stamped and kept; the model re-scores with the live values
Ask the Data	An AI assistant that writes database queries, reads the methodology and the 700 submissions, and can fetch nirfindia.org live. It displays the queries it ran so answers can be checked

Running it. `./run.sh`, then open `http://localhost:8501`. After NIRF 2026 is published, `./pipeline.sh` re-runs every step and this report can be rebuilt with `python docs/build_report.py`.

Folder structure

```
scraper/ scrape_rankings.py · download_pdfs.py · parse_pdfs.py · build_db.py – collect and store
models/ score_model.py (fits the f curves) · predict_2026.py (forecast) · artifacts/ (trained
model)
analysis/ analyze.py – cut-offs, tiers, movers, peers, gap → data/processed/analysis.json
app/ Home.py · common.py · pages/1-6 · rag/index.py (document search) · rag/chat.py (AI assistant)
data/raw/ html/ pdf/ pdf_text/ methodology/ saveetha/ – everything downloaded, untouched, kept for
audit
data/processed/ rankings.csv · submissions.csv · faculty.csv · analysis.json · prediction_2026.json
· model_report.json
db/nirf.db the one database file (SQLite); any SQL tool can open it
docs/ this report and the script that builds it · README.md · .env (API keys, never shared) ·
run.sh · pipeline.sh
```

11 Caveats

- The college's parameter scores for 2020-2025 are **model estimates**; NIRF has not published them. They are calibrated on 700 institute-years and the 2025 estimate agrees with the published band, but they are not official figures. The accurate phrasing is 'estimated at about 13', not 'our RPC is 13'.
- Publications, citations and Perception are not in the data. They are the main sources of uncertainty and are reported as ranges.
- The 2026 forecast assumes the 2025 methodology is unchanged. NIRF has changed weights and sub-parameters before (e.g. the 2025 retraction penalty); the pipeline re-reads the methodology PDF each year.
- The AI assistant can misstate wording. It displays the database queries it ran; figures that matter should be checked against the dashboard or this report.

12 Sources and evidence

Every figure in this report traces to one of the following. Raw downloads are kept unchanged under `data/raw/` so any number can be re-checked.

#	Source	Location	Used for
S1	NIRF Engineering ranking tables, 2017-2025	<a href="https://www.nirfindia.org/Rankings/<year>/EngineeringRanking.html">https://www.nirfindia.org/Rankings/<year>/EngineeringRanking.html (also Overall, College, University)	Ranks, totals, TLR/RPC/GO/OI/PR; cut-offs; tier averages; the college's 2017 and 2019 rows
S2	NIRF rank-band pages	.../EngineeringRanking150.html, ...200.html, ...300.html per year	The college's 2018, 2020, 2024, 2025 band placements
S3	NIRF participant lists	.../EngineeringRankingALL.html per year	Years the college applied
S4	Per-institute data PDFs (704)	<a href="https://www.nirfindia.org/nirfpdfcdn/<year>/pdf/Engineering/<institute id>.pdf">https://www.nirfindia.org/nirfpdfcdn/<year>/pdf/Engineering/<institute id>.pdf	Raw submissions of ranked institutes: model training data; top-100 medians
S5	NIRF methodology documents 2023-2025	<a href="https://www.nirfindia.org/nirfpdfcdn/<year>/framework/Engineering.pdf">https://www.nirfindia.org/nirfpdfcdn/<year>/framework/Engineering.pdf (and Overall.pdf, College.pdf)	Weights, sub-parameter marks, formulas quoted in section 2
S6	College's NIRF 2025 filings	https://saveetha.ac.in/nirf-documents-2024/ → Engineering-1.pdf, Innovation.pdf, SDG-1.pdf	2025 raw numbers and faculty list
S7	College's NIRF 2026 filing	https://saveetha.ac.in/wp-content/uploads/2026/02/SAVEETHA-ENGINEERING-COLLEGE20260216-.pdf	2026 raw numbers; basis of the 2026 estimate and forecast
S8	Score model output	data/processed/model_report.json (models/score_model.py)	Cross-validation results; the college's estimated parameter scores
S9	Forecast output	data/processed/prediction_2026.json (models/predict_2026.py)	Thresholds, band probabilities, lever values, top-100 model validation
S10	Analysis output	data/processed/analysis.json (analysis/analyze.py)	Tier profiles, raw-number medians, gap decomposition, Tamil Nadu list
S11	Database	db/nirf.db (scraper/build_db.py)	All of S1-S7 as tables; row counts in section 4

Source: db/nirf.db (all of the above, loaded by scraper/build_db.py). Report built by docs/build_report.py; every number is read from S8-S11 at build time, not typed in.